Local Credit, Global Settlement: Enhancing Trust-Based Credit Creation with On-Chain Reserve Contracts

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Abstract

In this work, we propose Basis, a monetary framework in which both local trust-based currencies as well as global blockchain-based trust-minimized ones can coexist via smart contract powered reserves needed when there is no trust. Coordination with blockchain is needed only during debt note redemption against a reserve contract. Payments (debt note transfers) are done offchain, so with low fees and no need to use low-security centralized blockchain-like systems with high throughput. For coordination and transparency of debt in the system, a tracker service is used, which cannot steal money from reserves. We provide analysis of trust-minimization regarding tracker activities.

Current offchain payment systems backed by on-chain reserves (Lightning Network [14], Cashu [1], Fedimint [6]) require full backing and so do not allow for credit creation. In contrast, Basis allows for credit creation with no requirements set for reserves at the protocol level. Then it can be used for community trading without using blockchain at all (and possibly, without Internet even, just by using a local mesh network), but when trust is insufficient to expand credit, on-chain reserves can be established and used. We expect that on a local scale trust-based economic relationships would dominate, and on a global scale (for example, when one community is doing trades with another) full debt coverage with reserves would be needed in most cases. The whole system will use limited in disconnected from real-world needs supply blockchain assets only where they are needed (there is no trust), while enabling monetary expansion whenever possible to do that without enforcement (so where pure peer-to-peer trust can be established), thus allowing to create a viable alternative to political money.

The means of payment is an IOU note, which is signed by the issuer and also by a witnessing tracker. Our design allows for debt transferability, if the issuer agrees: if peer A issued debt to peer B, and peer B wishes to transfer this debt (fully or partially) to peer C, peer A signs an updated note reducing the debt to B and a new note creating the debt to C. After the tracker updates its state, peer A owes C directly. We show that

this design allows for minimal trust to tracker service. A tracker service is committing its state on the blockchain periodically. If a tracker service ceases to exist, it is possible to redeem debt notes against on-chain reserves using this committed state. There could be multiple trackers around the world. We consider different tracker designs, from just centralized server, to federation and dedicated sidechains.

We provide implementation of the reserve contract as well as offchain clients (tracker server and example clients). We show an example of group trading over a mesh network with occasional Internet connection. Another example shows AI agent economies where autonomous agents create credit relationships for services.

Keywords: digital currencies, cryptocurrencies, blockchain, peer-to-peer money, trust-based credit, IOU notes

1 Introduction

Peer-to-peer monetary systems, such as ChainCash [2], enable decentralized (peer-to-peer) money creation through trust and collateral. While on-chain protocols provide transparency and censorship resistance, they often struggle with scalability and cost when every transaction must be recorded on a public blockchain. Off-chain alternatives, such as payment channels and Layer 2 solutions, improve scalability but introduce new trust assumptions, and also do not allow for expansion on top of blockchain assets.

By expansion, we mean creating new monetary instruments (such as credit, IOUs, or representative tokens) that exceed the value of the underlying locked blockchain collateral. This is needed because the emission schedules of decentralized blockchain assets like Bitcoin or Ergo are algorithmically predetermined and entirely disconnected from real-world economic demand. Their supply cannot expand or contract in response to productive activity, trade volumes, or credit needs, making them inherently inelastic. Centralized alternatives like USDT or USDC reintroduce the very trust assumptions that blockchain technology sought to eliminate, requiring users to rely on opaque reserves, auditing processes, and issuer solvency. There is a need to bridge this gap by enabling elastic credit creation backed by trust, and this trust should be minimized in nature, so as to be peer-to-peer, while still anchoring to on-chain reserves when trust is insufficient.

Existing offchain payment systems demonstrate different approaches to this problem but fall short of enabling credit expansion. The Lightning Network [14] operates through bidirectional payment channels secured by hash time-locked contracts, requiring participants to lock collateral on-chain proportional to their maximum channel capacity. Cashu [1] implements Chaumian blind signatures to provide privacy-preserving ecash tokens, but the mint must hold full reserves backing all issued tokens. Fedimint [6] uses federated custody with threshold signatures to improve upon single custodian models, yet still requires full collateralization and introduces trust in the federation members. All three systems fundamentally require full backing of offchain liabilities with on-chain assets, preventing any form of credit creation or monetary expansion.

In this work, we propose Basis, a monetary framework designed for peer-to-peer credit creation and settlement. Basis enables participants to issue IOU notes without mandatory collateral reserves, allowing elastic credit creation backed purely by trust when counterparty relationships allow it. On-chain reserves serve as optional settlement backing for inter-community or low-trust transactions, rather than as a prerequisite for all economic interactions. This flexibility allows Basis to support both purely trust-based local currencies and fully collateralized global settlement within a unified framework. The offchain protocol maintainer (the tracker) regularly updates its efficient and verifiable commitment to offchain debt relationships between economic agents, witnessing new notes by signing them, and periodically committing state digests to a public blockchain. This design allows for efficient note circulation offchain while preserving the security properties of on-chain redemption.

The Basis protocol comprises two main components: on-chain reserve contracts and an offchain tracker service. Reserve contracts are smart contracts deployed on a public blockchain that lock collateral and enforce redemption rules. They verify cryptographic proofs of debt, check signatures from both the debtor and the tracker, and release collateral to creditors upon valid redemption requests. The tracker service maintains an authenticated ledger of all debt relationships, signs newly created IOU notes to certify their inclusion in the system state, and periodically commits cryptographic summaries of this state to the blockchain. This commitment enables participants to verify debt claims and provides an emergency redemption path if the tracker becomes unavailable.

We provide open-source implementations of both the smart contracts and the tracker service [10, 9], which we describe in Section 3.

The paper is organized as follows. In Section 2 we describe the design of Basis in detail. Section 3 provides implementation details. Section 4 analyzes security properties of the tracker.

2 Design

We consider a fully flat peer-to-peer monetary system where any economic agent may freely join. Each agent may issue IOU notes, thus creating debt relationships with other agents. Note acceptance is always optional and at the discretion of the counterparty.

2.1 Note Creation

When an agent (the payer) wishes to create a debt obligation to another agent (the payee), it creates an IOU note containing the payer's signature, the payee's public key, the debt amount, and a timestamp. The note may be associated with an on-chain reserve contract that backs the debt, if such a reserve exists. For the payee to accept the note, it must contain a valid tracker signature certifying that the tracker has recorded the debt in its authenticated state. The payee verifies this signature along with the payer's signature and the note's contents before accepting the note as a valid claim against the payer.

2.2 Note Redemption

A payee holding a witnessed IOU note may redeem it against the payer's reserve at any time by interacting with on-chain smart contract only. To redeem, the payee presents the note along with both the payer's and tracker's signatures to the reserve contract. The contract verifies the signatures, checks the debt amount against the tracker's committed state, and releases the corresponding collateral to the payee. If the reserve has insufficient collateral, the payee may still hold the note as a claim and attempt redemption later, or the payer may be required to top up the reserve. It is also possible to redeem the note partly.

In the case where the tracker becomes unavailable, an emergency redemption path exists: after a timeout period (e.g., 3 days), the tracker signature becomes optional. The payee can redeem using only the payer's signature and the last committed tracker state from the blockchain, preventing funds from being locked indefinitely.

2.3 Debt Transfer

When an agent wishes to transfer a portion of its debt to a new creditor, the process resembles triangular trade. Suppose peer A owes peer B, and peer B wishes to transfer part of this debt to peer C. Peer B requests peer A to co-sign new notes: one reducing the debt A→B and another creating debt A→C. The tracker witnesses both the reduction in the old debt and the creation of the new debt, and the old note is replaced with a new one. After the transfer, peer A owes peer C directly. This enables debt restructuring and circular trade without requiring full redemption and re-issuance.

2.4 Tracker State

Basis tracks all debt relationships in an AVL tree [15] maintained by the tracker. The tree maps a composite key to the current debt state:

$$\text{hash}(\text{payerKey}||\text{payeeKey}) \rightarrow (\text{totalDebt}, \text{timestamp}),$$

where hash is a cryptographically strong hash function.

Periodically, the tracker commits the root digest of this AVL tree to the public blockchain. This commitment provides a timestamped checkpoint of the global debt state and enables emergency redemption when the tracker is unavailable.

3 Implementation

We provide implementation of the Basis protocol in form of executable code, including smart contracts for on-chain reserves, a tracker server for coordinating offchain state, and client libraries for issuing and managing IOU notes.

The smart contracts and documentation are available at [10], and the tracker server implementation is available at [9].

3.1 Reserve Contract

The reserve contract is implemented on the Ergo blockchain using the ErgoScript smart contract language [3, 4], and is available at [8]. It locks collateral (ERG or custom tokens) and allows redemption when presented with a valid debt note and signatures. The contract enforces the timeout rules for emergency redemption without tracker signatures.

3.2 Tracker Server

The tracker server is implemented in Rust [11] and maintains an in-memory authenticated tree of all debt relationships. When a payer creates a note, the tracker verifies it, updates its local state, and returns a signature certifying the debt. The tree stores cumulative debt amounts indexed by the hash of the payer and payee public keys, enabling efficient lookups and proof generation.

The server exposes an HTTP API for note creation, querying, and redemption preparation. It also monitors the Ergo blockchain for reserve boxes and periodically submits commitment transactions that embed the current state digest on-chain. This allows parties other than the tracker to verify debt claims, redeem them on-chain, and provides an emergency redemption path if the tracker goes offline.

3.3 Client Libraries

Client libraries provide functionality for agents to issue IOU notes, request tracker witnessing, transfer debt to new creditors, and redeem notes against reserves. The libraries handle signature generation, note serialization, and communication with tracker and blockchain nodes.

3.4 Mesh Network Trading

We demonstrate community trading over mesh networks using Meshtastic LoRa devices. In this scenario, Alice and Bob are in a village with no direct Internet connectivity, but a local tracker running on a community server has connectivity to the Ergo blockchain via a satellite or cellular uplink. Alice creates an IOU note for goods purchased and sends it to Bob over the mesh network (Bluetooth/WiFi Direct/LoRa). Bob forwards the note to the tracker through the same mesh, and the tracker—being online—verifies it, updates its local state, signs it, and commits the updated digest to the blockchain. Multiple transactions accumulate this way, with the tracker continuously updating the on-chain state. When Bob later wishes to redeem, the tracker provides the necessary proof from its committed state, and the redemption transaction is submitted directly to the blockchain. This demonstrates offline credit-based trading for end users without forced collateralization, while the tracker maintains blockchain connectivity for state commitments and settlement.

Notably, the tracker's Internet connectivity can in principle be limited or occasional, as it may operate in SPV [12] or NiPoPoW [7] client mode, syncing with the blockchain only periodically rather than maintaining a full node connection [5]. This makes the architecture suitable for regions with intermittent or censored connectivity.

3.5 AI Agent Credit Creation

We also demonstrate how Basis can be used in autonomous credit creation among AI agents in a software development workflow. Initially, agents interact purely on trust without any reserves. A repo agent posts a development task with a budget, a dev agent completes the work and submits a pull request, and a test agent reviews and tests the code. The repo agent creates an IOU note (e.g., 10 ERG) for the dev agent, and the dev agent transfers a portion (e.g., 3 ERG) to the test agent for testing services. All these transactions occur autonomously without human intervention and without any collateral backing.

After observing the agents' work, humans may manually review the output and choose to back the agents by depositing assets into on-chain reserves. Once reserves are established, the IOU notes become redeemable against collateral. This creates a two-phase economy: first, agents build reputation through pure trust-based credit creation, and then humans provide reserve backing as economic validation of the agents' productive work. By creating credit first and earning reserves later, AI agents essentially create money through their economic activity, and humans then decide how sound that money is based on whether they provide backing after reviewing the agents' swarm output.

4 Tracker Security Analysis

In this section we provide preliminary security analysis for the Basis offchain tracker.

4.1 Tracker's Trust Model

For all the tracker-related trust and security issues, the most natural and also easiest in our case solution is to reduce trust in tracker itself. The simplest option which is implemented at the moment is just a single trusted entity. The following options exist to improve the tracker service:

- Federated service: we may have signing key being shared among several parties. With Schnorr signature scheme being used, it can be done via MuSig2 protocol [13]. Blockchain commitment to the tracker state then can be protected by multi-signature lock as well. In principle, signing and state commitments can be protected by different committees even.

- Sidechain: to witness debt notes, a dedicated blockchain-based ledger can be used. In case of permissioned consensus trust assumptions are not different from a federated service. In case of permissionless Proof-of-Work consensus, censorship-resistance can bSecurity Analysise improved in principle. To avoid higher fees to be paid to miners, merged mined sidechain option can be used [16].

4.2 Tracker Cannot Steal Funds

The tracker only certifies inclusion of notes in its state. It cannot unilaterally create or redeem notes. The issuer's signature is always required for redemption, ensuring that the tracker cannot steal funds.

4.3 Emergency Exit

If the tracker becomes unavailable, an emergency exit mechanism activates after a timeout period (e.g., 3 days or 2,160 blocks). After this period, the tracker signature becomes optional for redemption. The same message format is used: key||totalDebt||timestamp. Replay attacks are prevented by timestamp verification, and the reserve owner's signature is still required.

4.4 Anti-Censorship

The tracker cannot steal but it can refuse to witness debt notes for some issuer or recipient at some point. The pseudonymous nature would allow a payment recipient to pick another public key easily. However, network level anonymization could be needed in this case. To avoid reserve owner censorship, the reserve contract has a refund option, which allows the reserve owner to withdraw funds without a tracker after a long enough time. If the tracker is implemented as a trusted server or federation, it may disallow redemption by omitting a note from committed data on-chain. In a sidechain implementation, this can be prevented through consensus rules, or by storing a commitment to all previously published commitments, which can be enforced through on-chain contract logic for commitment updates. Future research should investigate a possibility to use blind signatures and zero-knowledge proofs (of no-doublespend, and so on) to have the tracker sign a note without knowledge of sender and receiver (beyond what zero-knowledge proofs may reveal).

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